

Yuan Gao

Ph.D. Candidate & Research Associate — Autonomous Vehicle Systems
Technical University of Munich

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🔗 [yuangao-tum](https://yuangao-tum.github.io) | 🎓 [Google Scholar](https://scholar.google.com/citations?user=yuan-gao-tum) | [in](https://www.linkedin.com/in/yuan-gao-tum) [yuan-gao-tum](https://www.linkedin.com/in/yuan-gao-tum) | 📄 [ResearchGate](https://www.researchgate.net/profile/Yuan-Gao)



🎯 **Actively seeking a Ph.D. research internship** in foundation models for autonomous driving — foundation-model-based scenario generation & analysis, and vision-language-action (VLA) models.
Available from early 2027 for 3–6 months.

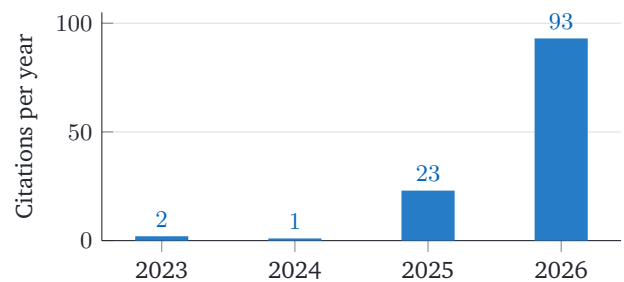
RESEARCH INTERESTS

Foundation models (LLMs, VLMs, VLAs, world models) for autonomous driving — scenario generation and scenario analysis, safety-critical scenario evaluation, visual question answering and benchmark construction, semantic anomaly detection, agent-level risk assessment; motion planning and optimal control of multi-agent systems.

RESEARCH IMPACT

- 📄 **121 citations** total | h-index 4 | i10-index 2
- 🔔 **93 citations in 2026 alone** — already ~4× the 2025 total
- ★ **228 GitHub stars** on the FM-AD-Survey repository
- 📄 **15 research outputs** (11 peer-reviewed) — 9 first-authored

Source: [Google Scholar](https://scholar.google.com/citations?user=yuan-gao-tum), August 2026.



2026 is year-to-date; Scholar per-year counts may not sum to the total.

EDUCATION

Ph.D. in Mechanical Engineering (Autonomous Vehicle Systems)

Technical University of Munich (TUM), School of Engineering and Design

Professorship of Autonomous Vehicle Systems; advisor: Prof. Dr.-Ing. Johannes Betz.

Research: scenario generation and analysis for autonomous driving based on foundation models.

Oct 2024 – present

M.Sc. Development, Production, and Management in Mechanical Engineering

Technical University of Munich (TUM)

Grade 1.5 — passed with distinction (double Master's program).

Oct 2021 – Aug 2023

M.Sc. Mechatronics and Robotics

Technical University of Munich (TUM)

Grade 1.2 — passed with high distinction.

Master's thesis at the Chair of Information-oriented Control (Prof. Dr.-Ing. Sandra Hirche, Dr. Junjie Jiao).

Oct 2019 – Feb 2023

M.Sc. studies, Mechatronics and Information Technology

Karlsruhe Institute of Technology (KIT)

Continued Master's studies at TUM from Oct 2019.

Oct 2018 – Sep 2019

B.Eng. Mechanical Design, Manufacture, and Automation

Hefei University of Technology, China

GPA 3.74/4.3, rank 3/384 — Excellent Graduate honors.

Exchange semester at I-Shou University, Taiwan (Feb – Jun 2015), GPA 4.0/4.0.

Sep 2013 – Jun 2017

AWARDS & HONORS

- Double M.Sc. passed with **high distinction** (Grade 1.2, Mechatronics & Robotics) and **distinction** (Grade 1.5), TUM.
- **Excellent Graduate** honors, Hefei University of Technology — undergraduate rank **3 / 384**.
- Perfect GPA (4.0 / 4.0) on exchange semester, I-Shou University, Taiwan.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Associate / Ph.D. Candidate

Oct 2024 – present

Professorship of Autonomous Vehicle Systems, TUM

- Lead research on foundation-model-based scenario generation and analysis for autonomous driving in a project funded by the German Central Innovation Programme for SMEs (ZIM).
- First-authored publications at ICRA, IROS, ITSC, and IEEE OJ-ITS; co-authored papers at ECCV (2×) and ITSC; built open-source frameworks, benchmarks, and datasets (NuRisk, FM-AD-Survey, From Words to Collisions).
- Supervised 13 student theses/projects; teaching assistant for the F1TENTH practical course and the Future Mobility Camp seminar.

Research Assistant

Jul 2023 – Nov 2023

Chair of Information-oriented Control, TUM

Distributed optimal control of multi-agent systems. Afterwards (Dec 2023 – Sep 2024), prepared Ph.D. applications and extended the Master's thesis results into a journal submission (IEEE TCNS, under review).

Intern — Healthcare / Medical Robotics

Feb 2023 – Jun 2023

ITK Engineering GmbH (Bosch Group)

Internship in the healthcare and medical robotics product area.

Student Research Assistant

Mar 2022 – Aug 2022

Chair of Automotive Technology, TUM (Prof. Dr.-Ing. Markus Lienkamp)

Driving-simulator research for automated driving; previously completed a semester thesis at the same chair (May – Dec 2021).

PUBLICATIONS

Author name in bold; citation counts from Google Scholar, August 2026. 9 of 15 publications are first-authored.

Journal Articles

- [J1] **Y. Gao**, M. Piccinini, Y. Zhang, D. Wang, K. Moller, R. Brusnicki, B. Zarrouki, A. Gambi, J. F. Totz, K. Storms, S. Peters, A. Stocco, B. Alrifae, M. Pavone, J. Betz. “Foundation Models in Autonomous Driving: A Survey on Scenario Generation and Scenario Analysis.” *IEEE Open Journal of Intelligent Transportation Systems*, 2026. **82 citations** [DOI] [arXiv] [code ★228]
- [J2] **Y. Gao**, J. Jiao, Z. Li, S. Hirche. “H₂ Suboptimal Containment Control of Homogeneous and Heterogeneous Multi-Agent Systems.” *IEEE Transactions on Control of Network Systems* (under review). [arXiv]

Conference & Workshop Papers

- [C1] **Y. Gao**, M. Piccinini, R. Brusnicki, Y. Zhang, J. Betz. “NuRisk: A Visual Question Answering Dataset for Agent-Level Risk Assessment in Autonomous Driving.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. **6 citations** [arXiv] [code ★7] [dataset]
- [C2] **Y. Gao***, W. Miao*, M. Piccinini, H. Wang, Q. Song, J. Betz (* equal contribution). “Chat2Scenic: An Iterative RAG-Based Framework for Scenario Generation in Autonomous Driving.” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026 (accepted).
- [C3] F. R. Schäfer, **Y. Gao**, D. Wang, T. Stauner, S. Günemann, M. Piccinini, S. Schmidt, J. Betz. “EgoDyn-Bench: Evaluating Ego-Motion Understanding in Vision-Centric Foundation Models for Autonomous Driving.” *European Conference on Computer Vision (ECCV)*, 2026 (accepted). [arXiv]
- [C4] D. Wang, Z. Liang, H. Ye, Z. Sun, Z. Lu, Y. Zhang, Y. Zhao, **Y. Gao**, M. Seegert, F. R. Schäfer, H. Qin, W. Li, L. Palmieri, F. Jahncke, M. Piccinini, J. Betz. “Target-Bench: Can Video World Models Achieve Mapless Path Planning with Semantic Targets?” *European Conference on Computer Vision (ECCV)*, 2026 (accepted). **6 citations** [arXiv] [code] [project] [dataset]
- [C5] R. Brusnicki, D. Pop, **Y. Gao**, M. Piccinini, J. Betz. “Can VLMs Unlock Semantic Anomaly Detection? A Framework for Structured Reasoning.” *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2026 (accepted). [arXiv] [code] [project]
- [C6] Q. Song, **Y. Gao**, R. Brusnicki, F. Sarro. “Warnless at the DeepTest 2026 Tool Competition.” *7th IEEE/ACM International Workshop on Deep Learning for Testing (DeepTest)*, 2026. [ACM DL]
- [C7] **Y. Gao**, M. Piccinini, J. Betz. “Towards Whole-Body VLA: A Scalable Data Collection Framework for Quadrupedal Mobile Manipulators.” *2nd German Robotics Conference*, 2026. [PDF]
- [C8] F. R. Schäfer, K. Moller, **Y. Gao**, C. Oefinger, S. Schmidt, J. Betz. “Imagined Rollouts are Kinematic, Not Dynamic: A Diagnosis of Long-Horizon World-Model Failure.” *RSS Robot World Model Workshop*, 2026. [arXiv]
- [C9] **Y. Gao**, M. Piccinini, K. Moller, A. Alanwar, J. Betz. “From Words to Collisions: LLM-Guided Evaluation and Adversarial Generation of Safety-Critical Driving Scenarios.” *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2025. **17 citations** [arXiv] [code ★31]

- [C10] **Y. Gao**, J. Jiao, S. Hirche. “H₂ Suboptimal Leader-Follower Consensus Control of Multi-Agent Systems.” *IFAC World Congress*, 2023. IFAC-PapersOnLine 56(2), 2614–2619. [3 citations](#) [\[code\]](#)
- [C11] **Y. Gao**, J. Jiao, S. Hirche. “H₂ Suboptimal Containment Control of Multi-Agent Systems.” *American Control Conference (ACC)*, 2023. [\[IEEE Xplore\]](#) [\[code\]](#)

Preprints

- [P1] Q. Song, **Y. Gao**, J. Betz, D. Pfahl, M. R. Mousavi, F. Sarro. “In the Driver’s Seat: A Multi-Company Study on the Reality of Autonomous Driving System Testing.” arXiv:2607.15820, 2026. [\[arXiv\]](#)
- [P2] Z. Li, F. Zhao, M. Piccinini, B. Zarrouki, **Y. Gao**, Z. Shan, J. Betz, C. Lv, L. Xie. “SGTP: Sampling-based Game-Theoretic Planning for Real-Time Multi-Vehicle Autonomous Racing.” arXiv:2607.25388, 2026. [\[arXiv\]](#) [\[code\]](#) [\[project\]](#)
- [P3] **Y. Gao**, D. Hua, M. Piccinini, F. R. Schäfer, K. Moller, L. Li, J. Betz. “StyleVLA: Driving Style-Aware Vision Language Action Model for Autonomous Driving.” arXiv:2603.09482, 2026. [3 citations](#) [\[arXiv\]](#)

TALKS & PRESENTATIONS

IEEE ICRA 2026 (Vienna, Austria) — presented *NuRisk* [C1]. Jun 2026

IEEE ITSC 2025 (Gold Coast, Australia) — presented *From Words to Collisions* [C8]; workshop talks at the *LLVM-AD* and *Interaction-driven Behavior Prediction & Planning* workshops. Nov 2025

Research visit, Australian Centre for Robotics, University of Sydney. Nov 2025

ACADEMIC SERVICE

Conference reviewer: NeurIPS 2026 • ICLR 2026 • CVPR 2026 • ICRA 2026, 2025 • IROS 2026 • ITSC 2026, 2025 • IEEE IV 2026, 2025 • IFAC World Congress 2023.

Journal reviewer: IEEE Open Journal of Intelligent Transportation Systems • IEEE Robotics and Automation Letters.

TEACHING (TECHNICAL UNIVERSITY OF MUNICH)

Teaching Assistant, “F1TENTH: Autonomous Driving Hands-on” (practical course, English). WS 2024, WS 2025

Teaching Assistant, “Future Mobility Camp” (project-week seminar, practical sessions). WS 2024, WS 2025

Tutor, “F1TENTH: Autonomous Driving Hands-on” (practical course, English). WS 2024

Tutor, “The Driving Simulator in the Development Process” (practical course, German). SS 2022

Certification: TUM Onboarding Program, elective module “Designing Courses.” Dec 2024

STUDENT SUPERVISION (13 THESES & PROJECTS)

- Yang Zhang, *Ethics-Aware Decision Modeling for Autonomous Driving Using Large Vision-Language Models*, Master’s thesis, Feb – Aug 2026
- Yuheng Liu, *CoVLM: A Vision-Language-Model-Based Semantic V2V Cooperation Agent for Autonomous Driving*, Master’s thesis, Feb – Aug 2026
- Chen Li, *Multimodal Retrieval-Augmented Closed-Loop Autonomous Driving Scenario Generation*, Master’s thesis, Jun – Dec 2025 – Jun 2026
- Joshua Pusch, *A VLM-Based Scenario Analysis of Autonomous Driving*, interdisciplinary project, Mar – Sep 2025 – Mar 2026
- Haoyu Wang, *Closing the Loop: Integrating Vision-Language Models for Feedback-Driven ADAS Scenario Generation*, Master’s thesis, Mar – Sep 2025 – Mar 2026
- Mengchen Cai, *From State Machines to Traffic Safety: Benchmarking Driver Assistance Systems Using Structured Scenario Models*, research internship, Aug 2025 – Feb 2026
- Karmishth Annauth, *Literature Study: Lightweight and Fast-Inference Vision-Language Models for Autonomous Driving*, semester thesis, Aug 2025 – Feb 2026
- Wenting Miao, *Retrieval-Augmented Generation of Simulator-Compatible ADAS Scenarios*, Master’s thesis, Jul 2025 – Jan 2026
- Dengyuan Hua, *VLM-based Trajectory Generation: Driving Behavior Diversity*, Master’s thesis, Jul 2025 – Jan 2026
- Yung Loon Lim, *Bridging ADAS/ADS Functionality, Regulations, and Scenario-Based Testing: A Comprehensive Literature Review*, semester thesis, Jun – Dec 2025
- Leonardo Zambrano, *Fine-Tuning Vision-Language Models for Risk Assessment in Autonomous Driving*, Bachelor’s thesis, Apr – Oct 2025
- Sebastian Müller, *Generation of Safety-Critical Scenarios Based on LLMs*, Bachelor’s thesis, Mar – Sep 2025
- Marek Lorenz, *Risk-Aware Scenario Analysis Based on LLMs in Autonomous Driving*, Bachelor’s thesis, Oct 2024 – Apr 2025

OPEN-SOURCE SOFTWARE

- [FM-AD-Survey](#) (★ 228) — curated paper collection on foundation models for scenario generation & analysis in autonomous driving.
- [From-Words-to-Collisions](#) (★ 31) — LLM-guided evaluation and adversarial generation of safety-critical scenarios (ITSC 2025).
- [IFAC2023](#) (★ 24) & [ACC2023](#) — MATLAB implementations of H₂ suboptimal multi-agent control papers.
- [Riskaware-Scenario-analyse](#) (★ 21) — risk-aware scenario analysis based on LLMs.

SKILLS

Programming	Python, C, C++, C#
ML / Foundation Models	PyTorch, Hugging Face, LLMs & VLMs, agentic AI (LangChain, LangGraph, RAG)
Robotics & Systems	ROS/ROS 2, Linux, Git, CARLA simulation
Modeling & CAD	MATLAB/Simulink, Modelica, SILAB, PLC (CODESYS); AutoCAD, CATIA, Creo, Solid-Works
Languages	Chinese (native) • German (fluent, C1/DSH-2) • English (fluent, C1/IELTS 7.0)